

Assignment #2

Designing a Scalable Big Data System: Storage, Processing & Tools

Course/Code: **Big Data System Design (MBAI 5110G)**

Semester/Year: **Winter-2026**

Professor/Instructor: **Dilli Prasad Sharma, PhD**

Submission Due: **February 27, 2026 (End of Day)**

Learning Objectives

Upon completion of this assignment, students will be able to:

- Analyze real-world big data scenarios to identify data characteristics such as volume, velocity, and variety.
- Design a large-scale, fault-tolerant big data system architecture suitable for enterprise applications.
- Select and justify appropriate Hadoop and Spark ecosystem tools for data ingestion, storage, and processing.
- Communicate and justify architectural decisions through clear technical documentation and system diagrams

Submission Instructions

All students **MUST** complete and submit this assignment individually via Canvas, uploading a **single PDF or Doc** containing the **report**. Your submitted file name should be like:

<LastName_StudentID_Assignment_2_Report.pdf>

Report Format

- Title page
- Table of Contents
- Introduction and Problem Statement
- Big Data Storage Design
- Big Data Processing Design
- Big Data Tools and Ecosystem
- Conclusion
- References (IEEE/APA style)

(You can have many subsections/subtopics under a section)

Scenario Description

Suppose you are appointed as a **Big Data Architect** for a large **e-commerce platform** that manages a massive and diverse set of data, including:

- Millions of daily transactions, such as order details, payments, and inventory updates
- Continuous user activity logs and clickstreams capture browsing behavior
- Frequent product catalog updates from multiple vendors
- Customer reviews and social media interactions

Design a **big data system** to efficiently store, process, and analyze e-commerce platform data for reporting, analytics, and actionable insights.

Assignment Tasks:

Your report should address the following key aspects, focusing specifically on Hadoop and Spark-based ecosystem:

1. Big Data Storage Design (30%)

- Identify the data types of the e-commerce platform
- Design a storage architecture using Hadoop Distributed File System (HDFS) and Hive for structured querying.
- Explain data partitioning, replication, and fault-tolerance in HDFS.
- Discuss when to use HBase for real-time read/write access and Hive/Parquet for analytics.
- Include a diagram showing data storage layers and interaction between HDFS, Hive, and HBase.

2. Big Data Processing Framework (30%)

- Identify processing needs: batch and real-time streaming
- Propose processing frameworks:
 - Apache Spark for batch and streaming analytics
 - Spark Structured Streaming or Apache Flink
 - Use MapReduce for legacy batch jobs where appropriate
- Explain how fault tolerance is achieved with Spark and HDFS integration.
- Include a data flow/block diagram showing ingestion → storage → processing → analytics pipeline.

3. Tools and Ecosystem (40%)

- Focus on Hadoop/Spark-based tools:
 - Data ingestion: Apache Flume or Apache Sqoop
 - Storage/management: HDFS, Hive, HBase (NoSQL)
 - Processing: Spark Core, Spark SQL, Spark Streaming, MapReduce
- Explain the integration and workflow of these tools within the system with a diagram.